Locomotion and Movement

Introduction and Types of Movements

1. Read the following statements and identify the incorrect statement:
1) Ciliary movement is observed in human trachea and female reproductive tract.
2) Hydra uses tentacles for its movements.
3) Paramoecium uses flagella for its locomotion and movement of food through cytopharynx.
4) Amoeboid movement is affected by pseudopodia formation due to streaming of cytoplasm.

2. Find the mismatch:
1) Leucocytes- Amoeboid movement
2) Spermatozoa-Ciliary movement
3) HydraTentacles
4) Euglena-Flagellar movement

3. Which type of movement does macrophages and leucocytes exhibit?
1) Gliding movement
2) Amoeboid movement
3) Ciliary movement
4) Flagellar movement.

4. Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): All locomotion's are movements, but all movements are not locomotion's.

Reason (R): Walking, running, climbing, and swimming are types of voluntary movements, also called locomotion

In the light of the above statements, choose the correct answer from the option given below.
1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

5. Which structure is involved in both movement and locomotion?
1) Limbs in human body
2) Tentacles in the Hydra
3) Cilia in the Paramaecium
4) All of these

6. Given below are two statements.

Statement I: Movement is one of the significant features of living beings - both plants and animals.

Statement II: Streaming of protoplasm in the unicellular organisms like Amoeba is a simple form of movement.

Choose the correct option.

In the light of the above statements, choose the correct answer from the option given below.
1) Both statements I and II are incorrect.
2) Statement I is correct, but statement II is incorrect.
3) Statement I is incorrect, but statement II is correct.
4) Both statements I and II are correct.

7. Given below are two statements.

Statement I: Locomotion requires a perfect coordinated activity of muscular, skeletal and neural systems.

Statement II: Based on their location, three types. of muscles are identified: skeletal, smooth and cardiac.

In the light of the above statements, choose the correct answer from the option given below.

1) Both statements I and II are incorrect.
2) Statement I is correct, but statement II is incorrect.
3) Statement I is incorrect, but statement II is correct.
4) Both statements I and II are correct.

Muscle

8. During contraction of skeletal muscle, which of the following does not occur?
1) The length of isotropic band shortens.
2) The length of anisotropic band shortens.
3) The distance between two successive Z-lines decreases.
4) The length of H-zone shortens.

9. Where is the ATPase of the muscle located?
1) Actin
2) Myosin
3) Troponin
4) Actin

10. Which of the following is true regarding thick filaments?
1) Many monomeric proteins called meromyosin constitute one thick filament.
2) Each meromyosin has two parts, a globular head with a short arm and a tail, the former being called the heavy meromyosin (HMM) and the latter called the light meromyosin (LMM).
3) The globular head is an active ATPase enzyme and has binding sites for ATP and active sites for actin.
4) All statements are true.

11. Which of the following refers to 'cross arm', in the myosin (thick) filament?
1) Light meromyosin (LMM)
2) Head and short arm projecting outwards at regular distance and angle from each other
3) Trimers of troponin
4) Tropomyosin and troponin

12. Read the following and mark the wrong one among them:
1) Skeletal muscles are classified as red and white muscles based on the amount of haemoglobin in them.
2) Repeated stimulation of muscles can cause accumulation of lactic acid in them.
3) Binding of myosin head with new ATP results in breakdown of cross bridges during muscle contraction.
4) During contraction of skeletal muscle, the size of I-band decreases while that of A-band remains same.

13. Cross bridge pulls the attached actin filaments towards the centre of
1) 'I' band
2) 'Z' line
3) 'A' band
4) All of the above

14. Identify the correct set of statements.
(a) Each myosin filament is a polymerised protein.
(b) Myoglobin content is high in red muscle fibres.

(c) Troponin is a filamentous protein which run close to the 'F' actin filaments.
(d) When new ATP binds with actin then cross bridge is broken.
Choose the correct answer from the options given below.
1) (a), (c) and (d)
2) (a), (b) and (d)
3) (a) and (b)
4) (c) and (d)

15. Which of the following is not the component of heavy meromyosin?
1) Tail
2) A globular head
3) Cross arm
4) Both (1) and (2)

16. Which of the following is incorrect w.r.t. skeletal muscle?
1) Striated
2) Voluntary
3) Unbranched
4) Do not get fatigued

17. A protein having both structural and enzymatic properties is
1) Actin
2) Myosin
3) Tropomyosin
4) Troponin

Skeletal System

18. Mark the mismatch:
1) Upper jaw-Maxilla
2) Lower jaw-Mandible
3) Atlas-Olecranon process
4) Scapula-Acromion process

19. Given below are two statements.
Statement I: Each rib is a thin, flat bone connected dorsally to vertebral column and ventrally to the sternum.
Statement II: Each rib has two articulation surfaces on its ventral end and is hence called bicephalic.
In the light of above statements, choose the correct answer from the option given below.
1) Both statements I and II are incorrect.
2) Statement I is correct, but statement II is incorrect.
3) Statement I is incorrect, but statement II is correct.
4) Both statements I and II are correct.

20. Match Column I with Column II and select the correct option.

	Column I		Column II
(i)	Carpal	A	Foot
(ii)	Tarsal	B	Wrist
(iii)	Cranium	C	Frontal
(iv)	Face	D	Zygomatic

1) (i)-B, (ii)-A, (iii)-C, (iv)-D
2) (i)-A, (ii)-B, (iii)-C, (iv)-D

3) (i)-D, (ii)-C, (iii)-B, (iv)-A
4) (i)-A, (ii)-D, (iii)-C, (iv)-B

21. Choose the mismatch w.r.t. total number of different bones in humans:
1) Tarsals: 14
2) Carpals: 16
3) Ear ossicles: 3
4) Thoracic vertebrae: 12

22. Mark the odd one w.r.t the functions of vertebral column:
1) Attachment to large muscles of back
2) Protection of spinal cord
3) Attachment to ribs
4) Attachment to sternum

23. Which of the following is correct w.r.t. axial skeleton?
1) Axial skeleton has bones distributed along the main axis of the body.
2) The skull, vertebral column, sternum and ribs constitute axial skeleton.
3) It comprises 80 bones.
4) All of the above

24. Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R)
Assertion (A): The 8th, 9th and 10th pairsofribs are called vertebrochondral ribs (false ribs).
Reason (R): These pairs of ribs do not articulate directly with the sternum but join the seventh rib with the help of hyaline cartilage.
In the light of the above statements, choose the correct answer from the option given below.
1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

25. Match the following columns and choose the correct option.

	Column I		Column II
1	Longest bone	a	Hyoid
2	Collar bone	b	Clavicle
3	U-shaped bone	c	Femur
4	Patella	d	Cup shaped

1) 1-b, 2-c, 3-d, 4-a
2) 1- c, 2-b, 3-d, 4-a
3) 1-c, 2-b, 3-a, 4-d
4) 1-b, 2-c, 3- a, 4-d

26. Ribs are attached to ____ .
1) Sternum
2) Scapula
3) Ilium
4) Clavicle

27. Identify the parts labelled A, B, C and D in the given diagram of human skull:

	A	B	C	D
1)	Parietal	Temporal	Frontal	Mandible
2)	Temporal	Occipital	Frontal	Maxilla
3)	Parietal	Temporal	Maxilla	Mandible
4)	Temporal	Parietal	Mandible	Maxilla

28. Which of the following statement is incorrect w.r.t. human skeletal system?
1) Bone and cartilage are specialized connective tissues
2) In humans, the skeletal system is made up of 206 bones and few cartilages.
3) It is grouped into two principal divisions, i.e., the axial and the appendicular skeleton.
4) Cervical vertebrae is present in back.

29. Which of the following is correct match w.r.t bone and its total number in human body?
1) Ribs: 12
2) Patella: 1
3) Cervical vertebrae : 10
4) Clavicle : 2

30. Read the following statements describing the girdle bones of human skeleton and identify the false one among them.
1) Pectoral and pelvic girdle bones help in the articulation of the upper and lower limbs respectively with the appendicular skeleton.
2) Each girdle is formed of two halves.
3) Each half of pectoral girdle consists of a clavicle and a scapula.
4) Pelvic girdle consists of two coxal bones.

31. A cup-shaped bone called A covers the knee B
Fill in the blanks with correct option describing A and B, respectively.
1) Patella, dorsally
2) Patella, ventrally
3) Talus, ventrally
4) Calcaneum, ventrally

32. The dorsal, flat, triangular body of scapula has a slightly elevated ridge called the A, which projects as a flat, expanded process called t B. The C articulates with this.
Fill in the blanks with correct options:

	A	B	C
1)	Spine	Coracoid	First rib
2)	Spine	Acromion	Clavicle
3)	Dens	Coracoid	Clavicle
4)	Coccyx	Acromion	First rib

33. Match the following columns and choose the correct option

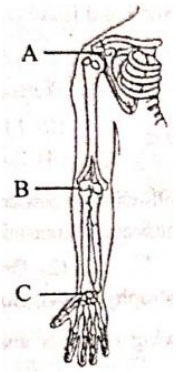
	Column I		Column II
1	Axial skeleton	a	14 in number
2	Appendicular skeleton	b	80 bones
3	Cranial bones	c	8 in number
4	Facial bones	d	126 bones

1) 1-b, 2-d, 3-c, 4- a
2) 1-d, 2-b, 3-c, 4- a
3) 1-b, 2-d, 3- a, 4-c
4) 1-d, 2-b, 3- a, 4-c

Joints, Disorders of Muscular and Skeletal System

34. Which one of the following is an example of immovable/ fibrous joint?
1) Sutures
2) Pivot joint
3) Both (1) and (2)
4) Hinge joint

35. In the given diagram, which joints are found at the marked sites:



	A	B	C
1)	Hinge	Ball and socket	Hinge
2)	Ball and socket	Hinge	Gliding
3)	Pivot	Hinge	Ball and socket
4)	Ball and socket	Pivot	Hinge

36. The joint between atlas and axis is
1) Saddle joint
2) Pivot joint
3) Hinge joint
4) Ball and socket joint

37. Choose the incorrect match:
1) Hinge joint-Knee joint
2) Pivot joint-Between humerus and ulna
3) Gliding joint-Between the carpals
4) Saddle joint--Between the carpal and metacarpal of thumb

38. Progressive degeneration of muscle mostly due to genetic disorder is called as
1) Myasthenia gravis
2) Osteoporosis
3) Muscular dystrophy
4) Tetany

39. Read the following statements:
(i) Joints are points of contact between bones or between bones and cartilages.
(ii) Force generated by muscles is used to carry out movement through joints, where muscles act as fulcrum.
(iii) Fibrous joints do not allow any movement.
(iv) In cartilaginous joints, the bones involved are joined together with the help of adipose tissue.
(v) Synovial joints are characterized by the presence of a fluid filled synovial cavity between the articulating surfaces of the two bones.

Which of the above statements are incorrect?
1) (i) and (ii)

2) (ii) and (iii)
3) (ii) and (iv)
4) (iii), (iv) and (v)

40. Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Tetany is rapid spasm in muscle.
Reason (R): Tetany is usually caused by an increase in the blood calcium level.
In the light of the above statements, choose the correct answer from the option given below.
1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

41. Match the following columns and choose the correct option.

	Column I		Column II
1	Fibrous joint	a	Gliding joint
2	Cartilaginous joint	b	Skull bones
3	Synovial joint	c	Joint between the adjacent vertebrae

1) 1-b, 2-c, 3-a
2) 1-c, 2-b, 3-a
3) 1-b, 2-a, 3-c
4) 1-a, 2-c, 3-b

42. Inflammation of joints is called
1) Arthritis
2) Tetany
3) Osteoporosis
4) Atherosclerosis

43. Which of following statements about gliding joints is incorrect?
1) These are flat joints.
2) Found between bones of carpals.
3) Twisting is not possible in these.
4) All of the above

44. Match the following columns and choose the correct option.

	Column I		Column II
1	Osteoporosis	a	Genetic disorder
2	Muscular dystrophy	b	Age-related disorder
3	Myasthenia gravis	c	Auto-immune disorder

1) 1-b, 2-c, 3-a
2) 1-c, 2-a, 3-b
3) 1-a, 2-b, 3-c
4) 1-b, 2-a, 3-c

45. Read the following statements and choose the incorrect option.
(i) Fibrous, hinge and pivot joints are different types of synovial joints.
(ii) Fibrous joints are present in the form of sutures, to form the cranium.
(iii) Synovial and cartilaginous joint allows considerable movement.
(iv) The joint between the adjacent vertebrae in the vertebral column is a cartilaginous joint.
Which of the above statements are incorrect?
1) (i) and (ii)
2) (i) and (iii)

3) (ii) and (iv)
4) (iii) and (iv)

46. Match the following columns.

	Column I		Column II
A	Freely movable joint	i	Synovial fluid
B	Sternum	ii	Flat bone
C	Cartilagenous joint	iii	Pectoral girdle
D	Glenoid cavity	iv	Vertebrae

1) A-ii, B-i, C-iv, D-iii
2) A-i, B–ii, C–iv, D-iii
3) A-i, B-iii, C-ii, D-iv
4) A-iv, B-iii, C-ii, D-i

47. Which of the following is not a disorder of bone?
1) Osteoporosis
2) Rickets
3) Atherosclerosis
4) Arthritis

48. Given below are two statements.
Statement I: Joints are essential for all types of movements involving the bony parts of the body.
Statement II: Joints are point of contact between bone and cartilages.
In the light of above statements, choose the correct answer from the option given below.
1) Both statements I and II are incorrect.
2) Statement I is correct, but statement II is incorrect.
3) Statement I is incorrect, but statement II is correct.
4) Both statements I and II are correct.

49. Which of the following disorder leads to fatigue, weakening and paralysis of skeletal muscle?
1) Osteoporosis
2) Myasthenia gravis
3) Gout
4) All of the above

50. Which of the following disorder is characterised by decreased bone mass and increased chances of fractures?
1) Tetany
2) Osteoporosis
3) Myotonic dystrophy
4) Gout

51. Match the following columns and choose the correct option.

	Column I		Column II
1	Tetany	a	Decreased levels of estrogen
2	Ball and Socket Joint	b	Cranium
3	Osteoporosis	c	Rapid spasms (wild contractions)
4	Fibrous joint	d	Humerus and pectoral girdle

1) 1-d, 2-c, 3-a, 4-b
2) 1-c, 2- d, 3-b, 4-a
3) 1-d, 2 – c, 3 - b, 4- a
4) 1-c, 2-d, 3- a, 4-b

Answer Key

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
3	2	2	1	4	4	4	2	2	4	2	1	3	3	1
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
4	2	3	2	1	3	4	4	4	2	1	3	4	4	1
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
2	2	1	1	2	2	2	3	3	2	1	1	4	4	2
46	**47**	**48**	**49**	**50**	**51**									
2	3	4	2	2	4									